Numerical Methods for Differential Equations in Model Rocket Simulation

Boula M. Etans

Riverside City College

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Abstract

Since the dawn of the space age, numerical methods have played a defining role in rocket development by enabling engineers to model trajectories long before physical testing. Because many governing equations in flight dynamics lack closed-form solutions (Karbon, 1998), numerical methods become the practical way to analyze and predict behavior. Building on these principles, this project applies those same numerical techniques to the study of small-scale rocket flight. This project extends a previously verified 1D ascent simulator into a full 3D Python-based flight simulation with explicit geometry, event-aware dynamics, and atmospheric modeling.

The rocket's geometry is defined component-wise (nose, body tubes, transitions, fins) with full parametric dimensions from which reference and wetted areas are computed; atmosphere modeling uses the International Standard Atmosphere (ISA) temperature and pressure, enabling Mach/Reynolds corrections, with angle-of-attack computed against a configurable wind model (validated here at zero wind, matching the actual launch conditions); aerodynamic forces follow a Barrowman-derived approach for normal force and center of pressure, including nose pressure drag, fin and body contributions, base drag, and transonic corrections consistent with model-rocket practice (Niskanen, 2013); 3D attitude is represented using quaternions while rotating aerodynamic and thrust vectors between body and inertial frames; translational and rotational dynamics are integrated using a fixed-stage fourth-order Runge-Kutta (RK4) scheme combined with adaptive step-size control, shrinking the timestep around ignition, burnout, and other rapidly changing force events and relaxing it during smoother coasting, selected over previously used Euler and classical predictor-corrector methods for better handling of discontinuities and rapidly changing forces during ascent (Derrick & Grossman, 1997; Bryan, 2025).

The prior 1D model developed for this project achieved agreement within 0.3% relative to OpenRocket predictions for a representative geometry, motivating its use as a baseline benchmark as the simulator expanded into 3D flight with attitude dynamics and off-axis forces; it has since been retired from the codebase now that the 3D model is independently validated. Validation of the new system will compare apogee, time histories of altitude, velocity, and acceleration against OpenRocket outputs and

flight-computer data from an actual model-rocket launch. The target accuracy is approximately 1-3% for major performance metrics under conditions where model assumptions remain valid. Beyond its computational contribution, this project shows how undergraduate students can apply numerical methods on differential equations to create end-to-end engineering simulations, turning abstract mathematics into practical tools and forming a foundation for future student research in rocket flight dynamics. In undergraduate engineering, numerical modeling has become especially important because it gives students an accessible means to explore complex real-world systems using mathematics and computation (Niskanen, 2013).

References

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